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AI-2002 Artificial Intelligence (AI)

Title : A Single-Agent AI System for Predicting Heart Disease Risk Using Artificial Neural Networks

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Abstract

HeartGuard is a single-agent AI system designed to predict the risk of heart disease in patients using an Artificial Neural Network (ANN). The system was trained on the UCI Heart Disease dataset, comprising 303 patient records with 13 clinical features. After data preprocessing, one-hot encoding of categorical variables, and standard scaling, the ANN model achieved 85% accuracy on a held-out test set of 60 patients. The model correctly identified 75% of true heart disease cases (recall) with a precision of 91%. The trained model is integrated into a prediction agent that accepts new patient data and returns a binary risk assessment. This report documents the complete development pipeline from data exploration to deployment.

1. Introduction

1.1 Background and Motivation

Cardiovascular disease remains one of the leading causes of mortality worldwide, accounting for approximately 17.9 million deaths annually according to the World Health Organization. Early and accurate detection of heart disease risk is critical , it allows patients to receive timely intervention, modify lifestyle factors, and avoid life-threatening cardiac events. However, traditional diagnosis relies on a clinician manually interpreting a combination of clinical measurements such as cholesterol levels, blood pressure, resting ECG results, and exercise-induced symptoms. This process is time-intensive, subject to human error, and not scalable to large populations.

Artificial Intelligence, and specifically Artificial Neural Networks, offers a compelling solution. ANNs can learn complex, non-linear relationships between multiple clinical variables simultaneously, relationships that may not be visible through standard statistical analysis. This motivates the development of an AI-powered prediction agent that automates the risk assessment process and provides clinicians and patients with a fast, data-driven second opinion.

1.2 Problem Statement

Diagnosing heart disease requires simultaneously evaluating over a dozen clinical parameters. Human analysis of these parameters at scale is difficult, and subtle interactions between variables , for example, the combined effect of chest pain type and maximum achievable heart rate , may go unnoticed. There is a need for an intelligent system that can automatically learn these relationships from historical patient data and apply them to predict risk in new patients accurately and consistently.

1.3 Objectives

The primary objectives of this project are:

- Develop a binary classification ANN to predict the presence or absence of heart disease.

- Preprocess the UCI Heart Disease dataset appropriately for neural network input.
- Evaluate model performance using accuracy, precision, recall, F1-score, and a confusion matrix.
- Integrate the trained model into a single intelligent agent capable of making predictions on new patient data.

1.4 Scope

This project is scoped as a supervised binary classification task. The dataset is limited to the Cleveland subset of the UCI Heart Disease repository (303 records). The model is not intended for clinical deployment but demonstrates the potential of AI-based decision support in a healthcare context.

2. Background

2.1 Artificial Neural Networks

An Artificial Neural Network is a computational model loosely inspired by the structure of biological neural networks. It consists of layers of interconnected nodes (neurons). Each neuron receives a weighted sum of its inputs, applies a non-linear activation function, and passes the output to the next layer. During training, the network adjusts its weights through a process called backpropagation, minimising the difference between its predictions and the actual labels using gradient descent.

For binary classification tasks, a sigmoid activation function is used in the output layer, producing a probability between 0 and 1. A threshold (typically 0.5) is then applied to produce a final binary prediction. The loss function used is binary cross-entropy, which penalises confident wrong predictions more heavily than uncertain ones.

2.2 Dataset

The UCI Heart Disease dataset (Cleveland subset) is a widely used benchmark in medical machine learning research. It contains 303 patient records collected at the Cleveland Clinic Foundation. Each record contains 13 clinical features and a target variable indicating the presence (1) or absence (0) of heart disease. The original target variable had values 0 through 4, which were binarised for this project. After removing 6 rows with missing values, 297 records were used for model development.

Key features include:

- **age** : patient age in years
- **cp** :chest pain type (4 categorical values: typical angina, atypical angina, non-anginal pain,

asymptomatic)

- **thalach** : maximum heart rate achieved during exercise
- **oldpeak** : ST depression induced by exercise
- **ca** : number of major vessels coloured by fluoroscopy (0–3)
- **thal** : thalassemia type (categorical: normal, fixed defect, reversable defect)

3. Methodology

3.1 Exploratory Data Analysis

Before any preprocessing, the dataset was explored to understand feature distributions, class balance, and relationships between variables. The target variable was found to be well-balanced , 54.1% of patients had no disease and 45.9% had disease , eliminating the need for resampling techniques such as SMOTE.

A Pearson correlation analysis was conducted to identify which features had the strongest linear relationship with the target. Key findings from the correlation analysis and visualisations are summarised below:

- thalach (max heart rate): strongest negative correlation with disease — disease patients had noticeably lower max heart rates, consistent with impaired cardiac function.
- cp (chest pain type): asymptomatic patients (cp=4) showed the highest disease prevalence — a known clinical phenomenon where the absence of pain does not indicate absence of disease.
- exang (exercise-induced angina): patients who experienced chest pain during exercise had significantly higher disease rates.
- ca (major vessels count) and oldpeak (ST depression): both showed positive correlation with disease.

Boxplots confirmed these findings visually, while a scatter matrix of continuous features (age, trestbps, chol, thalach, oldpeak) showed no severe multicollinearity between features.

3.2 Data Preprocessing

Missing Value Handling: Six rows containing missing values in the ca and thal columns were removed. Given the small dataset size, imputation was avoided to prevent introducing artificial patterns in medical measurements.

One-Hot Encoding: Four categorical features ; cp, restecg, slope, and thal were one-hot encoded. These features contain discrete unordered categories; treating them as raw integers would introduce a false mathematical ordering (implying, for example, that chest pain type 4 is twice as significant as type 2, which has no clinical basis). One-hot encoding converts each category into a separate binary column, expanding the feature space from 13 to 22 features.

Train-Test Split: The dataset was split 80/20 into training (237 records) and test (60 records) sets using stratified sampling to preserve the class balance in both subsets.

Feature Scaling: StandardScaler was applied to normalise all features to zero mean and unit variance. This step is essential for ANNs without it, features with large numerical ranges e.g., cholesterol: **126–564**, dominate the gradient update steps during training, preventing the network from learning meaningful weights for features with smaller ranges e.g., oldpeak: **0–6.2**. The scaler was fitted exclusively on training data and then applied to the test set, preventing data leakage.

3.3 ANN Architecture

A Sequential model was constructed using TensorFlow/Keras with the following architecture:

- Input layer: **22** features
- Hidden layer 1: **64** neurons, **ReLU** activation, followed by Dropout (rate=**0.3**)
- Hidden layer 2: **32** neurons, **ReLU** activation, followed by Dropout (rate=**0.3**)
- Output layer: **1** neuron, Sigmoid activation

The narrowing architecture (**64 → 32 → 1**) is intentional: the first layer extracts broad patterns across all **22** features, the second layer compresses these into stronger combined signals, and the output layer converts the final representation into a disease probability. Dropout layers randomly deactivate **30%** of neurons during each training step, preventing the model from over-relying on specific neurons and thereby reducing overfitting , a critical concern with small medical datasets.

The model contains **3,585** trainable parameters in total. This is deliberately modest given that the training set contains only **~237** patients; a larger model would memorise the training data rather than learn generalisable patterns.

3.4 Training

The model was compiled with the Adam optimiser and binary cross-entropy loss function. Adam was selected for its adaptive learning rate properties, which reduce the need for manual learning rate tuning. Training was configured for up to **200** epochs with a batch size of **16.**

An EarlyStopping callback monitored validation loss with a patience of **15** epochs. Training halted at epoch **18**, restoring the weights from epoch **3** , the point of lowest validation loss (**0.5691**). This outcome reflects a common challenge with small datasets: the model began overfitting after just **3** epochs, with training loss continuing to decrease while validation loss rose steadily. The EarlyStopping mechanism successfully prevented the model from training on this noise, automatically reverting to the optimal state.

4. Results and Evaluation

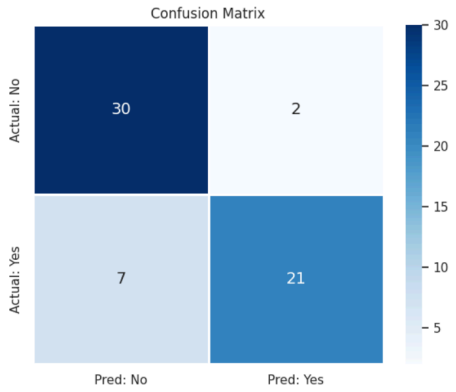
4.1 Classification Performance

The trained model was evaluated on the held-out test set of **60** patients. Results are summarised in the table below.

Class	Precision	Recall	F1-Score
No Disease	0.81	0.94	0.87
Disease	0.91	0.75	0.82
Overall Accuracy	—	—	0.85 (85%)

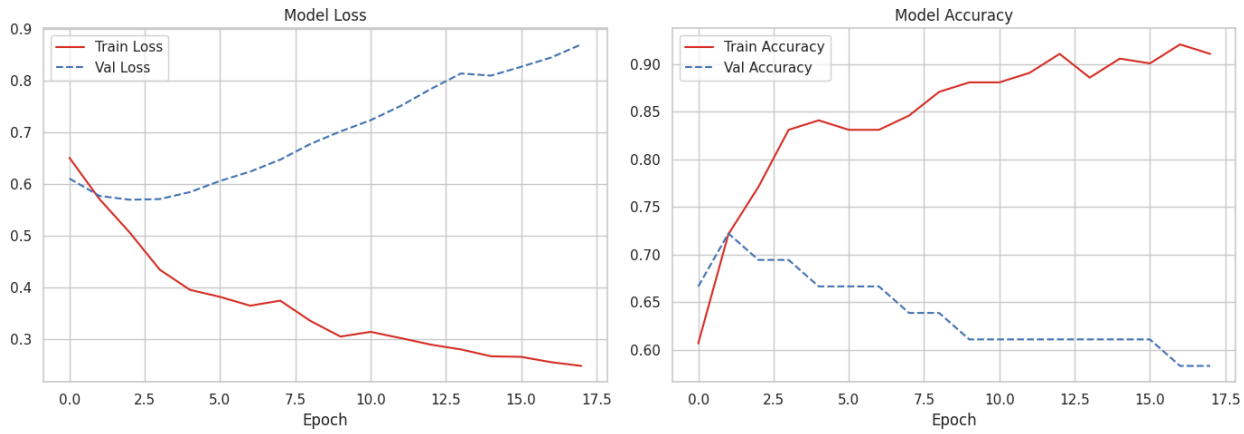
The model achieved **85%** overall accuracy. For the Disease class, precision of **0.91** indicates that when the model predicts heart disease, it is correct **91%** of the time. Recall of **0.75** indicates the model successfully identified **75%** of all true disease cases, missing **7** out of **28** diseased patients (False Negatives). In a clinical context, False Negatives are the more costly error ; a missed diagnosis could delay critical treatment. The **75%** recall is a known limitation of the dataset size.

4.2 Confusion Matrix



The confusion matrix shows: **30** True Negatives (healthy patients correctly identified), **21** True Positives (disease patients correctly identified), **2** False Positives (healthy patients incorrectly flagged), and **7** False Negatives (disease patients missed).

4.3 Training Curves



The training curves clearly illustrate the overfitting pattern: training loss decreased consistently across all **18** epochs, while validation loss reached its minimum at epoch **3** and subsequently increased. This divergence is the defining signature of overfitting and justified the use of EarlyStopping.

5. Conclusion

This project successfully developed HeartGuard, a single-agent AI system for binary heart disease risk prediction using an Artificial Neural Network. The complete pipeline , from raw data ingestion through exploratory analysis, preprocessing, model training, evaluation, and inference , was implemented in Python using TensorFlow/Keras, scikit-learn, and pandas.

The model achieved **85%** accuracy on unseen test data, with **91%** precision and **75%** recall for the disease class. While the recall limitation highlights the challenge of working with small medical datasets, the project demonstrates how AI agents can meaningfully assist in healthcare decision support by systematically applying learned patterns across multiple clinical variables simultaneously , a task that is difficult to perform consistently through manual analysis alone.

References

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